

A Proposed Nonlinear Programming Optimization Model for Optimal Budget Mix of Digital Marketing Campaigns

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Abstract—This paper proposes a nonlinear mathematical model to address the problem of optimizing budget allocation within digital marketing campaigns. The model provides marketers with a systematic quantitative framework for effective budget allocation across various platforms. The model seeks to identify the optimal budget mix that maximizes the expected net impressions using the Sainsbury Normal Method (SNM). The proposed model considers multiple constraints, such as budget limitations and advertiser's preferences. Results demonstrate the applicability of the model through hypothetical two case studies and scenario analysis. Furthermore, the model allows marketers to incorporate real-time data and adjust parameters based on the ongoing campaign's characteristics, audience, and advertiser's preferences.

Keywords—Marketing Budget Optimization, Nonlinear Programming Optimization, Sainsbury Normal Method, Digital Marketing, Social Media Platforms, Facebook/Meta, TikTok, LinkedIn.

I. INTRODUCTION

Digitalization has become a critical factor for business success in today's dynamic and competitive environment [1]-[2]. Consumers spend a significant amount of time engaging with content on various social media platforms [3]-[4]. Reflecting this trend, digital advertising spendings continue to rise, reaching \$601 billion in 2023 and projected to reach \$667 billion by 2024 with like 10% yearly increase [5].

Digital marketing encompasses advertising a business's products or services online through various digital channels, including smartphones, display ads, and platforms like Facebook and Google Ads [6]. Ultimately, the goal of digital advertising is to identify the optimal audience for a campaign, ensuring effective communication and achieving advertiser objectives [7]. Many companies leverage digital marketing to gain a competitive advantage. It fosters brand awareness and loyalty, increases website traffic, reduces marketing costs, and improves search engine ranking across various channels [8]-[11].

Digital marketing is a cost-effective and fast-paced strategy for gaining a competitive edge [12]. Its primary goals are to increase conversions and brand exposure through various platforms depending on the target audience and their preferred channels. Social media advertising spendings are also on the rise, projected to reach \$219 billion by 2024, from \$207 billion in 2023 with like 6% yearly increase [5].

Popular platforms include Facebook, Instagram, YouTube, TikTok, and LinkedIn. Their ability to reach a wider geographic audience, target specific demographics at

lower costs, and generate a higher return on investment makes online advertising attractive [13]. Such platforms lead user engagement and are crucial for any marketing strategy.

This study aims to answer the following two research questions: (1) What is the most effective way to allocate the digital marketing budget across different social media platforms (e.g., Facebook, Instagram, LinkedIn, and TikTok)? (2) How can the digital marketing budget mix be adjusted and optimized to be more agile and responsive to changing business sector characteristics and advertiser preferences?

This paper introduces a mixed-integer nonlinear optimization model. This approach empowers marketers to make informed digital marketing budget allocation within promotional campaigns. The proposed mathematical model aims to maximize the brand awareness, while considering budgetary and preferences constraints.

The remainder of this paper is organized as follows: Section 2 highlights the most related work. Section 3 provides the proposed optimization model. Section 4 presents two case studies used in this research. Section 5 examines the case studies and presents results, comparisons and discussions; and lastly, a conclusion is drawn in section 6.

II. RELATED WORK

In the era of booming digital advertising, data-driven budgeting and operational decisions are essential for optimal results. Applied Operations Research and Optimization Models are pivotal in optimizing promotion campaigns. Existing research explores various optimization models to improve medium selection and budget allocation for advertising campaigns. Bhatia *et al.* [14] used a linear programming model to maximize ROI by considering individual media ROI and budget constraints. Pferschy *et al.* [15] focused on media mix optimization using a Quadratic Knapsack Model that analyzes past campaign data and prioritizes recent performance. Usta *et al.* [16] discussed guidelines for developing marketing mix models for effective budget allocation, highlighting the need for further research on model validation and long-term optimization. Dorokhova *et al.* [17] introduced a system dynamics model that simulates consumer behavior over time to analyze resource allocation and long-term campaign impacts. These studies showcase the diverse approaches utilized to optimize campaign performance and resource allocation in marketing.

Nobibon *et al.* [18]-[19] developed models for selecting customer sets to receive product offers during promotions, considering business constraints and objectives. They

implemented branch-and-price algorithm, dynamic programming algorithm, and heuristics for larger campaigns. Furthermore, their work discussed the trade-off between solution quality and computation time. Hosein *et al.* [20] used a mathematical optimization model to determine the optimal customer targeting strategy for telemarketing, aiming to maximize success rate while minimizing customer inconvenience. Their results showed that the proposed method could enhance telemarketing success with a limited call budget. These studies highlight the effectiveness of optimization and data-driven approaches in improving marketing campaign resource allocation.

Most past literature has focused on general optimization models without serious validating effort through specific real case studies. Additionally, the allocation and distribution of marketing campaign's budget across digital platforms have not been extensively addressed. This study contributes to the literature in three key ways: First, it develops a mathematical model for the digital marketing budgeting mix problem. Second, it tests two weighting mechanisms for different digital marketing platforms based on target audience characteristics and advertiser preferences. Third, it provides two artificial case studies for model validation and experimental analysis.

III. THE PROPOSED MATHEMATICAL MODEL

This section presents the developed Mixed Integer Non-Linear Programming (MINLP) model for the budgeting mix optimization of digital marketing campaigns. The proposed model solves the digital marketing campaigns budgeting mix optimization problem. The objective function aims to maximize the expected net impressions, while adhering to budgetary and preferences constraints. Mathematical annotations used for the proposed MINLP mathematical model are summarized in Table 1, where i is an index for social media platforms/medium.

TABLE I. SUMMARY OF INDICES, PARAMETERS AND DECISION VARIABLES OF THE PROPOSED MODEL

Parameters	
	Number of intended planned days of the social marketing campaign for medium i .
	Average expected gross impressions per each dollar allocated for advertising in medium i .
	Lower bound of daily budget limit in US dollars could be assigned for medium i per day.
	Upper bound of daily budget limit in US dollars could be assigned for medium i per day.
	Weighting factor for medium i to prioritize and consider preferences of the advertiser.
	The total budget limit for the social marketing campaign.
	Expected total potential users could be reached in medium i .
Main Decision Variables	
	Amount or Budget in US Dollars assigned or dedicated for medium i per day.
Auxiliary Decision Variables	
	Expected gross impressions generated by medium i .
	Expected gross impressions percentage achieved in medium i .
	The total budget amount in US Dollars assigned or dedicated for medium i over the marketing campaign.

The objective function aims to maximize the Expected Net Impressions using the Sainsbury Normal Method (SNM) that is formulated mathematically in (1).

$$\text{Maximize } Z = \left[1 - \prod_i (1 - RP_i) \right] \quad (1)$$

Multi-channel social marketing campaigns face a difficulty of audience overlap. Ideally, each channel reaches a unique audience, but channels often share viewers. The Sainsbury Normal Method (SNM) [21], tackles this by providing a more accurate estimate of a campaign's net impressions [22]. This is crucial for effective budget allocation and campaign optimization.

The constraints in the proposed MINLP model ensure that the budget is allocated effectively across various digital advertising mediums while adhering to budgetary and preferences constraints. Following is the set of model constraints and a detailed description of them.

$$R_i = D_i * x_i * GRD_i \quad \forall i \quad (2)$$

$$RP_i = \frac{R_i}{PU_i} \quad \forall i \quad (3)$$

$$UB_i \geq x_i \geq LB_i \quad \forall i \quad (4)$$

$$B_i = D_i * x_i \quad \forall i \quad (5)$$

$$B_i \leq W_i * B \quad \forall i \quad (6)$$

$$\sum_i B_i \leq B \quad (7)$$

$$\sum_i W_i = 1 \quad (8)$$

$$x_i, B_i, R_i \geq 0 \text{ and } \in \mathbb{Z} \quad \forall i \quad (9)$$

Equation 2 calculates the expected gross impressions for medium i . It does this by multiplying the number of planned days (D_i), the daily budget allocated to medium i (x_i), and the average expected gross impressions per dollar spent advertising in medium i (GRD_i). Equation 3 then calculates the expected gross impressions percentage achieved in medium i through dividing the expected gross impressions (R_i) by the expected total potential users (PU_i) that might be reached in medium i .

Equation 4 ensures the daily budget (x_i) allocated to medium i falls within a predefined range, with a lower bound (LB_i) and an upper bound (UB_i). Equation 5 calculates the total campaign budget allocated to medium i (B_i) through multiplying the number of planned days (D_i) by the daily budget allocated to medium i (x_i). Equation 6 incorporates advertiser preferences into the budget allocation for medium i . It ensures the total budget allocated (B_i) doesn't exceed a nominal budget based on a weighting factor (W_i) and the total campaign budget (B). This weighting factor allows advertisers to prioritize channels they believe will be more relevant and effective. Equation 7 ensures the total budget allocated across all mediums doesn't exceed the overall campaign budget (B). Furthermore, Equation 8 guarantees that the sum of all weighting factors (W_i) equals 1. Finally, (9) enforces non-negative integer values for budgets and reach.

IV. CASE STUDY

This section presents two case studies based on artificial digital marketing campaigns. Through application of the model to those two case studies, authors aim to validate the model's effectiveness and its ability to generate an optimal

budget allocation that aligns with campaign goals and achieves desired outcomes.

In digital marketing campaigns using social media platform, costs vary based on the chosen platform, target audience, campaign impressions, and advertiser credibility. The study examines frequently used platforms like Facebook, Instagram (both under Meta Platforms), TikTok, and LinkedIn. Two sectors are investigated: Education and Fast-Moving Consumer Goods (FMCG). The Education sector has a diverse audience with varying backgrounds, including students, parents, and educators who actively use social media. Similar dynamics apply to the FMCG sector, which has a wide and varied audience across multiple segments.

For Meta Platforms in the education sector, the target audience were on average 35 million people, aged between 18-65+, with an interest in Egyptian education. The minimum daily budget was \$1, with an average expected reach of 5,050 impressions per dollar. For the FMCG sector, the estimated audience size was on average 20.5 million people aged between 18-55+. The minimum daily budget remained \$1, with an average expected reach of 5,000 impressions per dollar.

For TikTok, the target audience in the education sector were on average 20 million people aged between 18-55+. The minimum daily budget was \$10, while an average estimated reach of 1,750 impressions per dollar. The FMCG audience on TikTok were estimated on average at 23 million people aged between 18-55+, with a minimum daily budget of \$10 and an average estimated reach 2,500 impressions per dollar.

LinkedIn is only considered for the education sector due to its user base being less interested in FMCG compared to other platforms. The target audience in education was over 360,000 individuals, with a minimum daily budget of \$10 and an average expected reach of 160 impressions per dollar.

The total budget for the social media marketing campaign is assumed to be \$1000 for five days. Advertisers should assign weights to each platform, where these weights reflect advertiser's preferences, industry type, and target audience needs.

V. NUMERICAL RESULTS

The proposed model is solved optimally with the Excel Optimization Solver (GRG Nonlinear Engine) for the two different cases: Education Sector and FMCG Sector. Each case is solved under different instances of the overall campaign budget (B) and the weighting factors (W_i). This post-solution analysis tests how the model handles practical considerations like advertiser preferences and the campaign budget.

A. Case 1: Education Sector

This case study addresses a digital marketing campaign for an education institution (e.g., Nile University) that aims to achieve the highest impressions within the dedicated campaign budget. Three digital marketing channels are considered: Facebook and Instagram (Meta), TikTok, and LinkedIn. The case study and model performance is tested under three budgetary scenarios: Low Budget (\$500), Medium Budget (\$1000), and High Budget (\$1500). Furthermore, advertiser preferences and channels weighting factor effect is studied by testing the model and the resulted campaign budget mix under two preference scenarios: Balanced (the three

weighting factors are equally - $W_1 = W_2 = W_3 = 0.333$) and Adjusted (Higher weight is given to Meta according to the advertiser preferences and efficiency considerations ($W_1 = 0.5$, $W_2 = 0.3$, $W_3 = 0.2$)). Table II summarizes the key input parameters used for case 1.

Table III summarizes case 1 results under the medium budget and adjusted preferences scenario. Results show that most of the budget (50%) is allocated to META, with a daily budget of \$100 and a significant reach of 2,525,000 impression (82%) to the campaign. This indicates META's effectiveness in reaching a broad audience within the allocated budget. TikTok receives a daily budget of \$60, accounting for 30% of the total campaign budget, while TikTok brings in 17% of the total reach (525,000 impressions). While it does not reach as many people as META, it still contributes significantly to the overall reach. LinkedIn has the lowest budget allocation (20%) with a daily budget of \$40 while it has the least reach (32,000 impressions only 1%) among the three platforms.

Figure 1 depicts the overall impressions achieved through the two different budget allocation strategies (Balanced and Adjusted) across three different overall campaign budgets (\$500, \$1000, \$1500). On the one hand, both strategies show a positive relationship between the campaign budget and overall impressions. As the overall campaign budget increases, the overall impressions increase proportionally. On the other hand, the Adjusted strategy consistently outperforms the Balanced strategy at all budget levels. This suggests that the adjusted preferences better target the audience or allocate resources more effectively compared to the balanced approach. Furthermore, the increase in impressions with the Adjusted strategy is more significant at each step of the budget increase. For example, increasing the budget from \$500 to \$1000 results in an increase of 1,541,000 and 1,160,000 impressions for the Adjusted and Balanced strategy, respectively. This indicates that the Adjusted strategy not only starts at a higher baseline but also scales more effectively with increased budgets.

TABLE II. CASE 1 KEY INPUT PARAMETERS

	GRD	W	B	PU
META	5,050	0.50	500	35,000,000
TikTok	1,750	0.30	300	20,000,000
LinkedIn	160	0.20	200	360,000

TABLE III. CASE 1 SOLUTION SUMMARY UNDER MEDIUM BUDGET AND ADJUSTED PREFERENCES

	X	B	R	Budget %	Impressions %
META	100	500	2,525,000	50	82
TikTok	60	300	525,000	30	17
LinkedIn	40	200	32,000	20	1
Total		1000	3,082,000	100	100

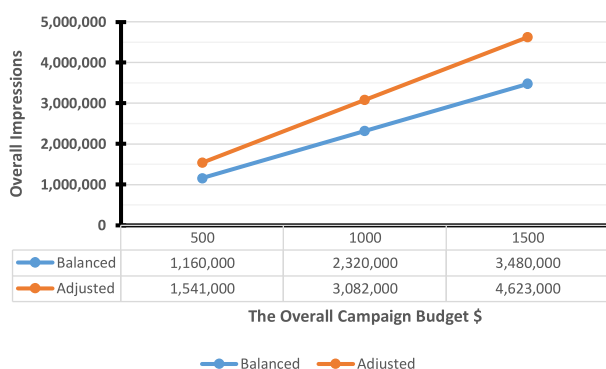


Fig. 1. Estimated impressions under different budgeting strategies and limits for case study 1

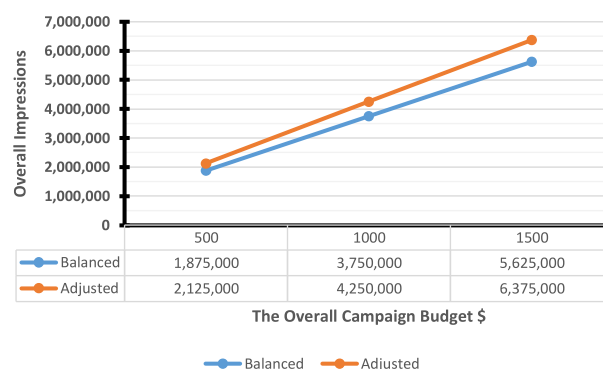


Fig. 2. Estimated impressions under different budgeting strategies and limits for case study 2

B. Case 2: FMCG Sector

This case study addresses a digital marketing campaign for a FMCG brand (e.g., Lamar Egypt) that aims to achieve the highest impressions within the dedicated campaign budget. For this case study, only two digital marketing channels are considered: Meta and TikTok. Similarly, the three budgetary scenarios are considered, and two preference strategies are tested: Balanced (the two weighting factors are equally - $W_1=W_2=0.5$) and Adjusted (Higher weight is given to Meta according to the advertiser preferences and efficiency considerations ($W_1=0.7$, $W_2=0.3$)). Table IV summarizes the key input parameters used for case 2.

Table V summarizes case 2 results under the medium budget and adjusted preferences scenario. Results show that most of the budget (70%) is allocated to META, with a daily budget of \$140 and a significant reach of 3,500,000 impression (82%) to the campaign. Similarly, META is dominant in reaching a broad audience within the allocated budget. TikTok receives a daily budget of \$60, and accounts for 30% of the total campaign budget, while TikTok brings in 18% of the total reach (750,000 impressions).

Like case 1, both strategies show a positive relationship between the campaign budget and overall impressions (Fig. 2). As the overall campaign budget increases, the overall impressions increase proportionally. Furthermore, Fig. 2 reinforces the superiority of the Adjusted strategy over the Balanced strategy and a clear advantage in terms of overall impressions.

TABLE IV. CASE 2 KEY INPUT PARAMETERS

	GRD	W	B	PU
META	5,000	0.70	700	20,500,000
TikTok	2,500	0.30	300	23,000,000

TABLE V. CASE 2 SOLUTION SUMMARY UNDER MEDIUM BUDGET AND ADJUSTED PREFERENCES

	X	B	R	Budget %	Impressions %
META	140	700	3,500,000	70	82
TikTok	60	300	750,000	30	18
Total		1000	4,250,000	100	100

The proposed model is not limited to the specific case studies discussed earlier in this paper but has broader applicability and can be adapted to different marketing environments, regardless of the specific platforms or industries involved, due to its generalizable structure. The model's adaptability is reinforced by its reliance on real-time data inputs, which allow marketers from diverse sectors to adjust parameters based on the unique characteristics of their campaigns.

VI. CONCLUSION

This study introduced a nonlinear mathematical model for optimizing budget allocation in digital marketing campaigns, specially on social media platforms. The model provided a quantitative framework to maximize net impressions while adhering to budget constraints and advertiser preferences. Through two hypothetical case studies, the results showed that the overall impressions increase proportionally with the overall campaign budget increases. Furthermore, Adjusted budget allocation strategy consistently outperforms the Balanced strategy across various budget levels, demonstrating greater efficiency in converting budget into impressions. The proposed model offers a robust and adaptable tool for brand awareness maximization within budget constraints, making it an asset for marketers. Future research should focus on refining the model and validating it through real-world case studies.

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